MODERN DEVOPS

3 CORE PRACTICES

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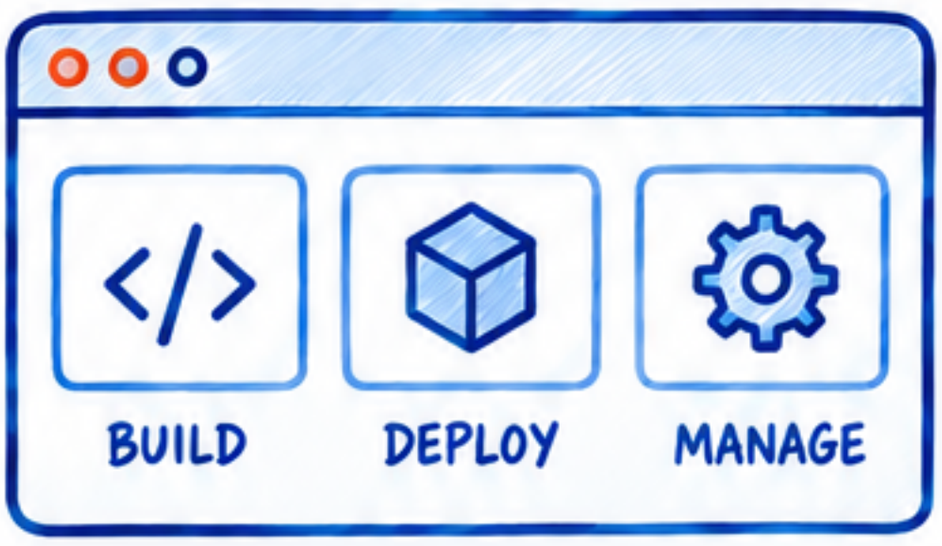
1. Rollback & Recovery

Restore services safely after failure.



2. Internal Developer Platforms

Provide standardized self-service workflows.



3. Policy Enforcement

Keep systems secure and compliant.



Together: Speed + Safety + Reliability

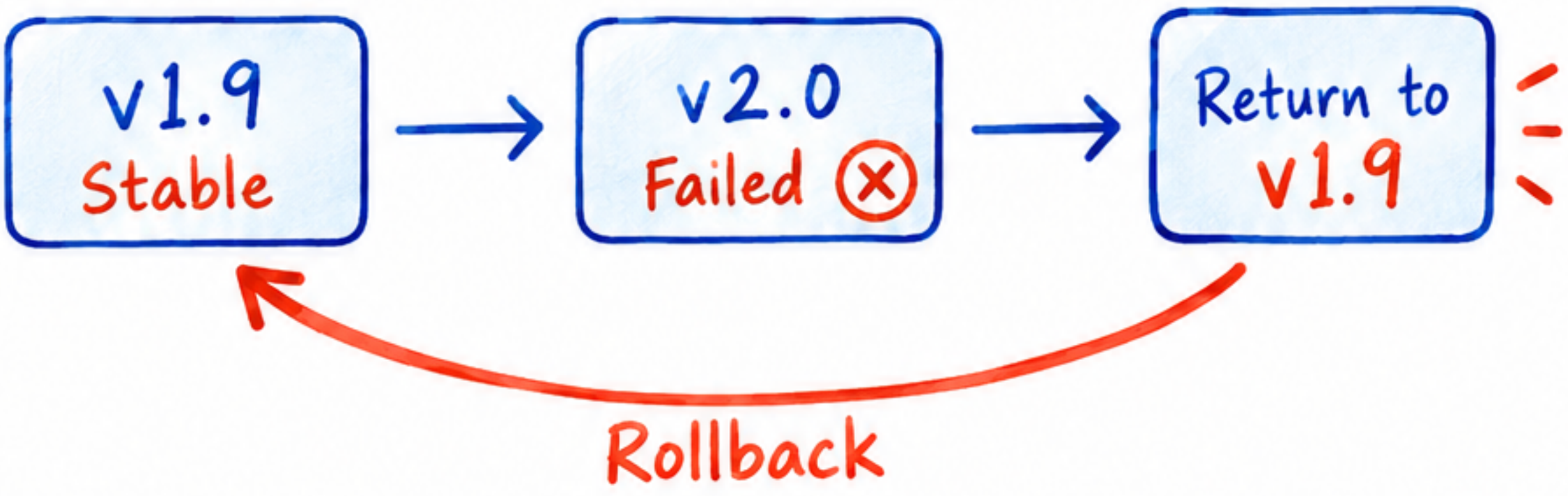
ROLLBACK

RETURN TO A STABLE VERSION

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Rollback means returning to a previously working application or configuration version.

Scenario:



Common methods:

- Redeploy the previous artifact
- Restore the previous configuration
- Shift traffic using blue-green deployment
- Disable a faulty feature with a feature flag

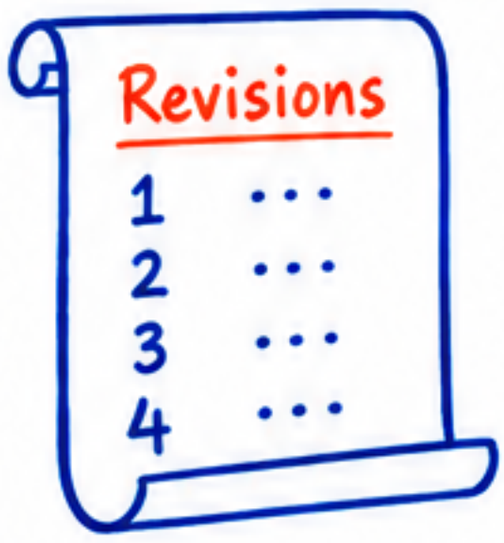
> ⚠ Database changes may not be safely reversible.

KUBERNETES ROLLBACK DEPLOYMENT COMMANDS

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1 View history

```
kubectl rollout history deployment/my-app
```



2 Undo deployment

```
kubectl rollout undo deployment/my-app
```



3 Check status

```
kubectl rollout status deployment/my-app
```

```
deployment/my-app
Waiting...
Progressing...
Successfully rolled out.
```

Mini workflow:



Note:
A rollback works best when images and manifests are versioned.

RECOVERY

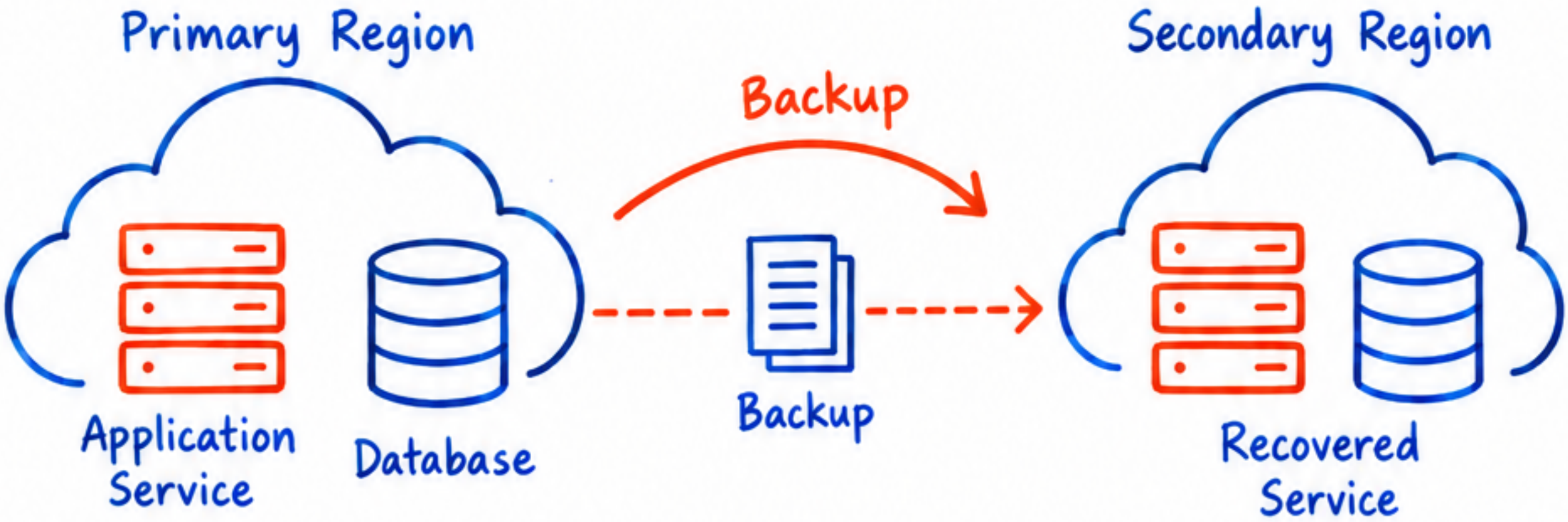
RESTORE SERVICE AND DATA

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Recovery restores the complete service after a serious failure or outage.

Examples:

- Restore a database from backup
- Rebuild infrastructure using Terraform
- Recover workloads in another cluster
- Restore persistent volumes
- Fail over to another region



Checklist:



- Backups tested ✓
- Runbook ready ✓
- Recovery drill completed ✓

RTO AND RPO
RECOVERY OBJECTIVES

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RTO – Recovery Time Objective

- Maximum acceptable time to restore service.
- Example: RTO = 30 minutes



RTO
Restore service within target time.

RPO – Recovery Point Objective

- Maximum acceptable data loss measured in time.
- Example: RPO = 5 minutes



QUICK COMPARISON

RTO asks: How quickly?

RPO asks: How much data?

INTERNAL DEVELOPER PLATFORM

STANDARDIZED SELF-SERVICE

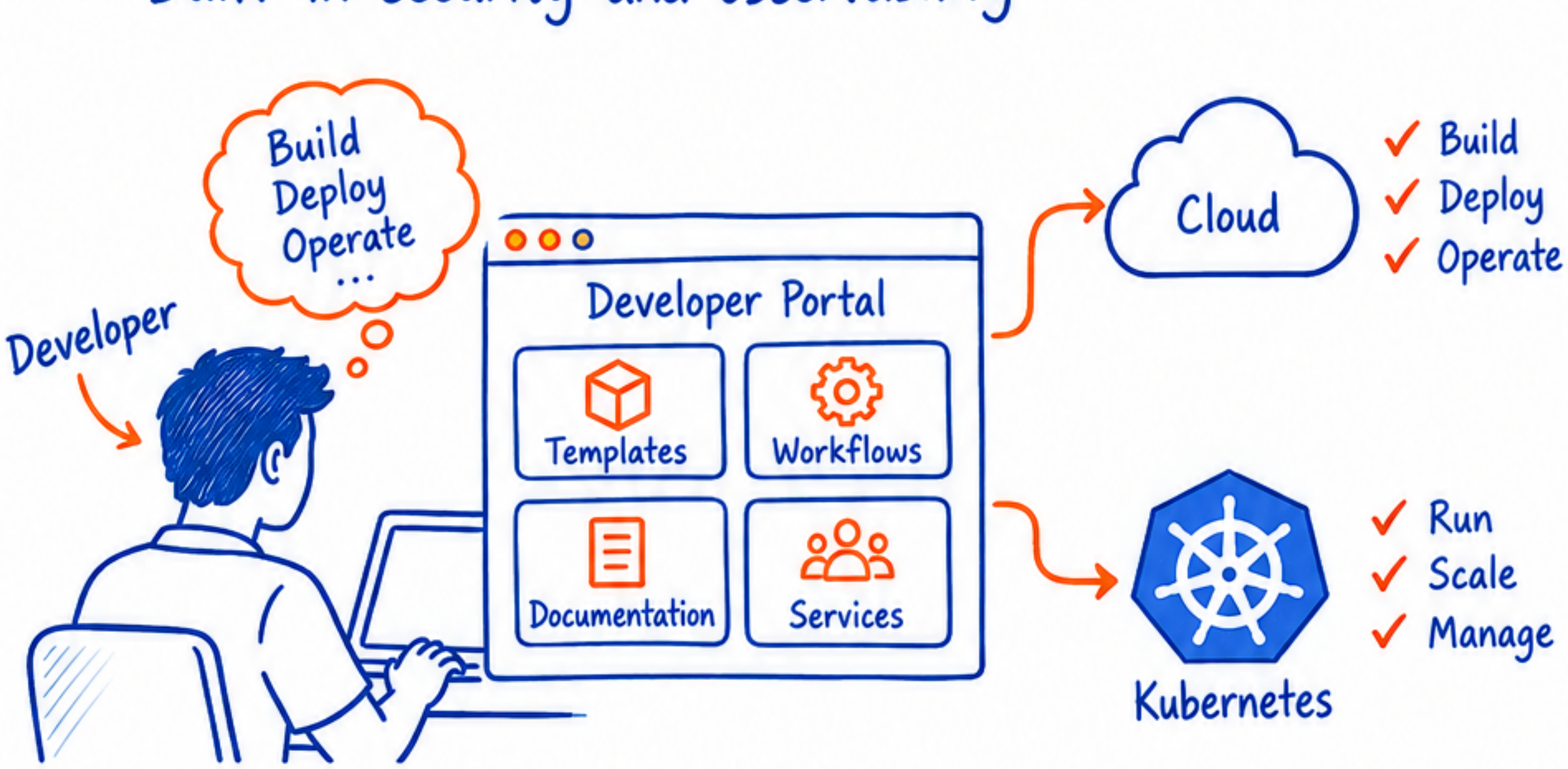
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Definition:

An IDP helps developers build, deploy and operate applications through approved self-service workflows.

Benefits:

- Faster onboarding
- Consistent deployments
- Reduced cognitive load
- Built-in security and observability



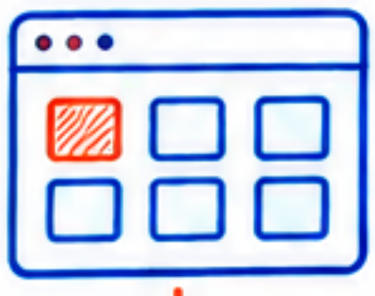
Key principle:

Treat the platform as an internal product.

IDP WORKFLOW

FROM IDEA TO PRODUCTION

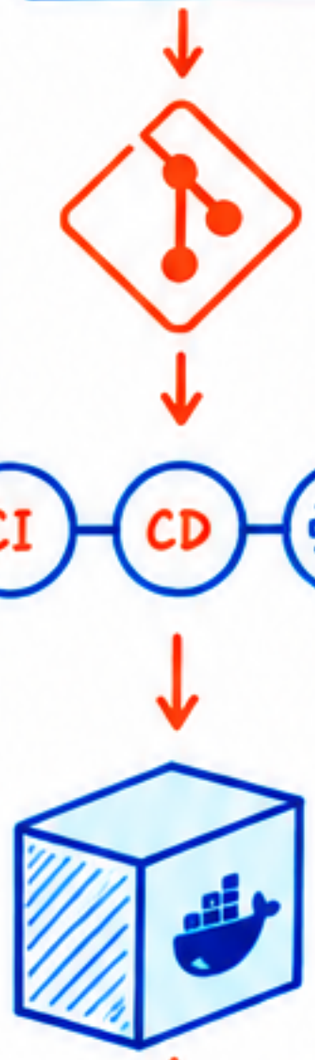
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1 Choose a service template

Start with a pre-defined template for your use case.

- Web app
- API service
- Background worker
- and more...



2 Create a Git repository

A new repository is created for your service code.

- Source control
- Branching strategy
- Initial commit



3 Generate a CI/CD pipeline

Automatically set up build, test and deploy pipelines.

- Build
- Test
- Security checks
- Deploy

4 Build a container image

Package your application into a container image.

- Dockerfile
- Build image
- Push to registry



5 Provision infrastructure

Set up required cloud resources using Infrastructure as Code.

- Compute
- Networking
- Storage
- IAM



6 Deploy to Kubernetes

Deploy your container to a Kubernetes cluster.

- Apply manifests
- Set up services
- Configure scaling



7 Enable logs and monitoring

Set up logging, metrics and alerts for observability.

- Centralized logs
- Metrics & dashboards
- Alerts



8 Register ownership and documentation

Set ownership, add metadata and generate documentation.

- Owner team
- Service catalog
- Runbooks
- Auto-generated docs



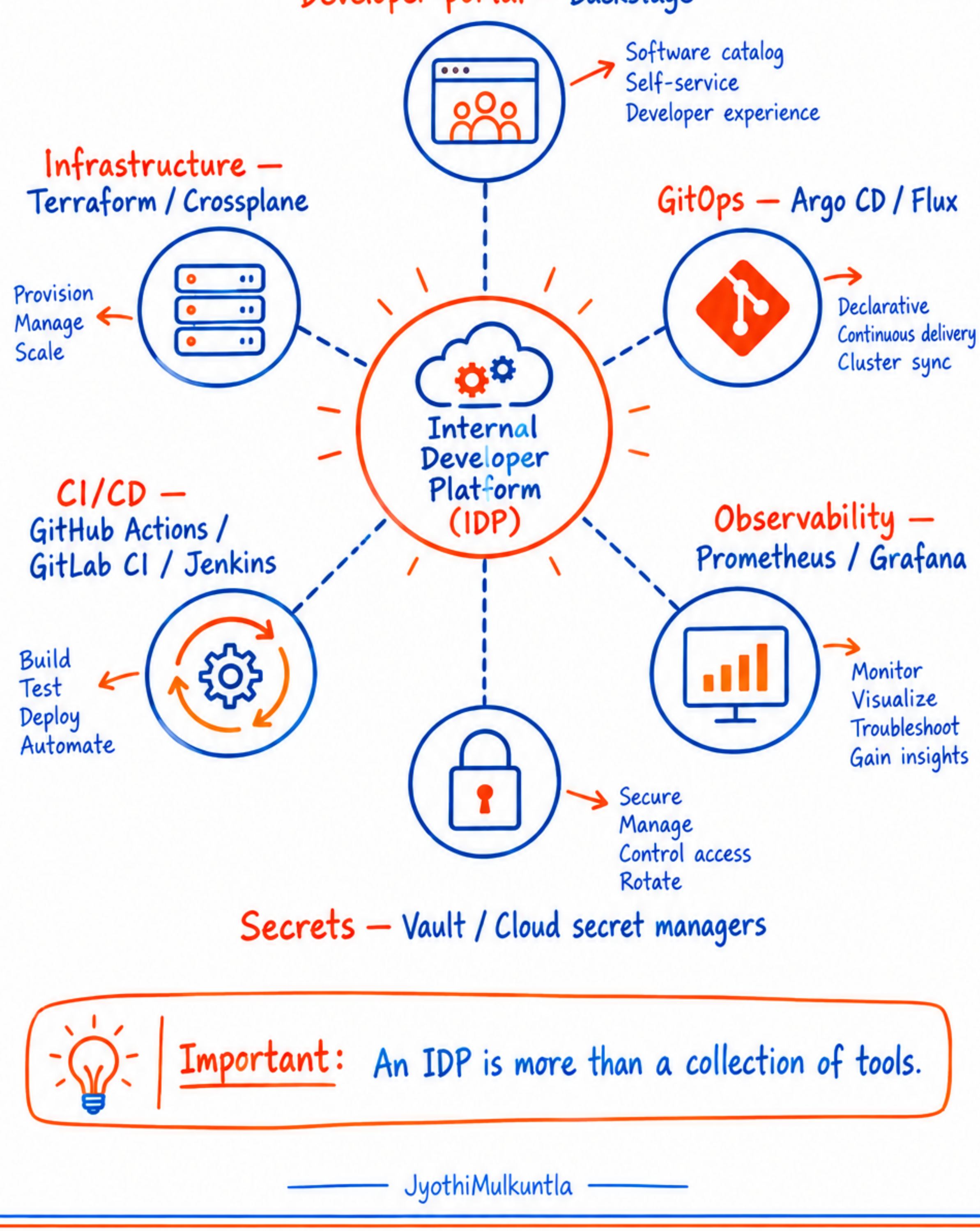
A repeatable golden path for developers

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PLATFORM CAPABILITIES

TOOLS THAT WORK TOGETHER

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Important: An IDP is more than a collection of tools.

POLICY ENFORCEMENT

SECURITY AS CODE

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Definition:

Policy enforcement checks that applications and infrastructure follow organizational rules.

Example rules:

- Containers must not run as root
- Privileged containers are prohibited
- Use approved image registries
- Define CPU and memory limits
- Add required labels
- Deploy signed images

Policy checks a Kubernetes manifest

pod.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: web
  labels:
    app: web
spec:
  containers:
  - name: nginx
    image: nginx:1.25
    securityContext:
      runAsNonRoot: true
    resources:
      limits:
        cpu: "500m"
        memory: "512Mi"
```



- Policy Compliant
- Allowed Registry
- Non-root User
- Resource Limits
- Required Labels
- Signed Image

Blocked unsafe container



```
image: untrusted:latest
securityContext:
  runAsUser: 0   # root
  privileged: true
```



BLOCKED

Policy Violation

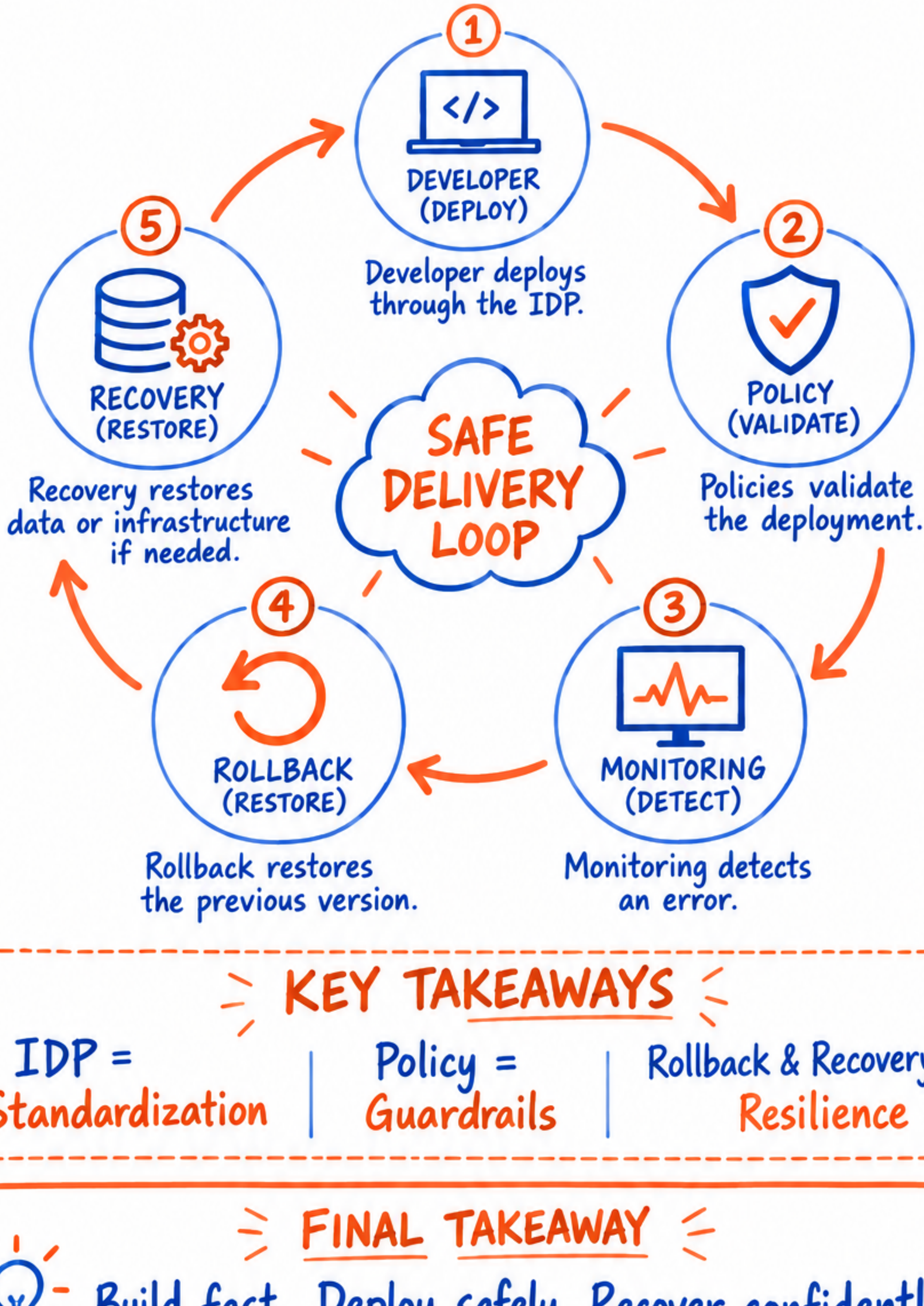
Modes:



HOW THEY WORK TOGETHER

SAFE DELIVERY LOOP

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KEY TAKEAWAYS

IDP = Standardization	Policy = Guardrails	Rollback & Recovery = Resilience

FINAL TAKEAWAY

Build fast. Deploy safely. Recover confidently.